

■# FLCR-34 - Electron-Hole-Exciton Accounting

Series: Formalizing LCR, Unifying Standard Models **Artifact:** formal paper source **Status:** first-pass rich rewrite; requires final citation and build pass. **Claim posture:** maximal local claim posture with explicit lane boundaries.

Abstract

This paper formalizes electron-hole-exciton formalism as standard-model-adjacent accounting for residues. The operative object is exciton accounting. The core result is that standard electron, hole, recombination, screening, and exciton models can explain part of the open residue surface when material context is declared. The paper also defines how this result routes forward: FLCR-34 supplies the missing condensed-matter bridge for papers that need carrier-pair accounting. Its main residue is explicit: material-specific bands, lifetimes, mobilities, and spectra require data-bound calibration.

Keywords

exciton accounting; LCR; receipt; claim lane; normal form.

External Reader Orientation

An outside reviewer should read this paper as a translation and comparator paper. Its local object is **Electron-Hole-Exciton Accounting**. The paper's immediate contribution is: Asks how much open CQE math is standard electron-hole-exciton formalism.

The nearest external anchors are standard mathematical physics definitions, cited Standard Model targets, calibrated constants or data where available, and the LCR-native papers that provide the source-side object. LCR terms are therefore internal labels for the construction unless a row explicitly imports an external theorem, citation, dataset, or calibration target.

It may close the external target definition and the internal translation grammar, but it does not claim measured physical identity unless a calibration/comparator row is present.

Reviewer Compression

Core object. Electron-Hole-Exciton Accounting.

Primary result. This paper contributes a controlled mapping between a cited external target and a cited LCR-native source object.

Primary non-result. This paper does not claim measured physical identity unless the comparator/calibration row is attached.

Review strategy. Evaluate the formal claim rows first, then inspect the receipt/citation/calibration rows that support each claim. Appendices provide routing and implementation context; they should not be read as stronger than their lane permits.

Peer Reviewer Assessment

Reviewer assumption: the reviewer has been briefed on the FLCR structure, claim lanes, CAM/crystal routing, forge/action surfaces, and the distinction between internal formal results and external physical calibration.

Recommendation. major revision, technically promising.

Review score vector. orientation=2, claim_granularity=2, evidence_visibility=2, falsifier_visibility=2, journal_apparatus=1, external_target_boundary=2.

Formal claim rows reviewed. 3.

Reviewer Strength Finding

The manuscript correctly treats Standard Model language as an external target surface and keeps translation claims separate from measured physical identity.

Major Revision Requests

- Convert the strongest proof sketch into numbered definitions, lemmas, and propositions with explicit dependencies.
- Replace placeholder source classes with final citation keys, receipt hashes, or dataset identifiers wherever possible.

- Move repeated pass-derived material into a compact evidence appendix and keep the main body focused on the paper-local argument.
- Add a comparator table mapping each external target object to the exact LCR source object, invariant, calibration status, and falsifier.

Minor Revision Requests

- Normalize theorem capitalization and punctuation.
- Add final figure/table captions where the text currently describes diagrams schematically.
- Ensure every acronym and local term appears in the paper-local glossary.
- State scale, scheme, and units for any physics-facing numerical comparator.

Acceptance Blockers

- Measured physical identity cannot be accepted without comparator/calibration rows.
- The mapping from external target to LCR source must be explicit enough that another reviewer could reject or reproduce it.

1. Contribution And Scope

- Defines exciton accounting as a first-class FLCR object.
- States the local result: standard electron, hole, recombination, screening, and exciton models can explain part of the open residue surface when material context is declared.
- Routes downstream use through claim lanes rather than inherited prose: FLCR-34 supplies the missing condensed-matter bridge for papers that need carrier-pair accounting.
- Preserves the residue boundary: material-specific bands, lifetimes, mobilities, and spectra require data-bound calibration.

This paper belongs to the Standard Model translation band. Standard Model language names an external target surface, not an automatic LCR identity. A translation claim is admissible only when the cited standard object, the LCR-native source object, the mapping rule, the evidence lane, and the falsifier or calibration boundary are all visible.

2. Source Routing

Primary routed sources:

- NP-12_Electron_Hole_Exciton_Accounting_For_Open_Math.md
- supplements/SM_BRIDGE_SUPPLEMENT.md

Cross-corpus evidence classes:

- CAM_CLAIM_100_SCORE_AUDIT.md
- NSIT_TOOL_CLOSURE_MATRIX.md
- NSIT_REASONED_CLOSURE_PASS.md
- PAPER_REWORKS_CRYSTAL_PROJECTION.json
- standards, glue, window, and node databases where applicable
- notebook-derived review prompts and orbital source manifests

3. Definitions

- **Hole.** The standard quasiparticle absence state in a band model.
- **Exciton.** A bound electron-hole excitation requiring material-specific parameters.
- **Receipt boundary.** The exact scope in which the paper's result can be replayed or consumed.
- **Promotion route.** The evidence path required before a stronger downstream claim can use this result.

4. Formal Claims

Claim	Lane	Statement
Theorem 34.1	standard_theorem_citation_bound_result	standard electron, hole, recombination, screening, and exciton models can explain part of the open residue surface when material context is declared
Proposition 34.2	normal_form_result	exciton accounting can be imported by later papers only through its declared source, receipt, and lane.
Protocol 34.3	falsifier_or_open_obligation	material-specific bands, lifetimes, mobilities, and spectra require data-bound calibration

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 34.1	external target or comparator	standard electron, hole, recombination, screening, and exciton models can explain part of the open residue surface when material context is declared	named external theorem, definition layer, citation target, or imported standard formalism	imports the external object as-is; does not silently reprove or redefine it	attach final citation metadata and mapping rule
Proposition 34.2	translation grammar	exciton accounting can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 34.3	obligation/falsifier	material-specific bands, lifetimes, mobilities, and spectra require data-bound calibration	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-34 defines or uses Electron-Hole-Exciton Accounting as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	The Standard Model or adjacent physics object is closed only as an external target definition when the relevant definition/citation row is present.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Measured values, physical identity, and predictive accuracy require scale, units, dataset, uncertainty, comparator, and pass/fail criteria.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

5. Paper-Specific Development

1. Identify the local object and its assigned sources for FLCR-34.

- 2. Separate what is internally receipt-bound from what is citation-bound, CAM-bound, calibration-bound, or still an obligation.
- 3. State the strongest admissible claim in its current lane.
- 4. Export only the lane-safe result to downstream papers and preserve the residue.

Proof Sketch

The proof strategy for FLCR-34 is a typed construction argument. The paper names hole as the active object, binds it to the routed legacy and orbital sources, and then allows downstream use only through the declared claim lane. This does not erase stronger ambitions; it keeps them addressable as dependencies, calibration tasks, or falsifiers.

6. Construction

The construction is intentionally staged. First, the paper names the finite or typed object that can be inspected directly. Second, it declares the admissible operations over that object. Third, it records the receipt or source class that allows another paper to consume the result. Fourth, it names the residue that must not be silently promoted.

For this paper, the operative object is Electron-Hole-Exciton Accounting. Its safe consumption rule is:

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source object -> declared operation -> receipt/lane -> downstream import
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The unsafe consumption rule is:

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source phrase -> downstream identity claim
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That second pattern is explicitly rejected by FLCR-01 and must be rewritten as a claim-lane promotion with evidence.

7. Evidence And Receipts

Evidence source	What it contributes
Legacy paper body	Primary formal and source-integration material assigned by the series map.
Internal and pyramid supplements	Cross-paper reasoning windows, deferred back-application slots, and route constraints.
NSIT tool closure matrix	Existing verifier/tool families that can close internal or batch claims.
CAM/crystal and standards-window evidence	Projected memory, source routing, standards reports, and window/node databases.

Standard Model Definition Dependencies

This paper consumes the dedicated Standard Model Defining layer. These papers define the external Standard Model or adjacent standard-physics object before this FLCR paper translates it into LCR language.

SMD paper	Definition surface	Required use
SMD -01	Standard Model Definition Contract And Evidence Boundary	Definition surface that must be cited before this paper promotes the corresponding Standard Model-facing claim.
SMD -07	Higgs Sector, Vacuum Structure, And Mass Terms	Definition surface that must be cited before this paper promotes the corresponding Standard Model-facing claim.
SMD -10	Electron Hole Exciton And Condensed Matter Correspondence	Definition surface that must be cited before this paper promotes the corresponding Standard Model-facing claim.
SMD -12	Standard Model Comparator And Falsifier Protocol	Definition surface that must be cited before this paper promotes the corresponding Standard Model-facing claim.

This paper also imports SMB-01, the Standard Model Closure Bridge. SMB-01 supplies the closure-by-lane rule: rows are no longer treated as unresolved by default; they are closed when standard formalism, FLCR proof, CAM/crystal reapplication, normal-form translation, or external calibration satisfies the lane, and they remain obligations only when a required field is missing.

Closed Rows Applied In This Paper

Row	Closure	Evidence	Boundary
Hole as effective vacancy carrier	closed by standard condensed-matter formalism	SMD-10, exciton source registry	Closes the meaning of hole inside band/effective theory.
Exciton as electron-hole bound state	closed by standard formalism	SMD-10	Closes vocabulary and model class, not material-specific values.
LCR vacancy/residue analogy	closed as normal-form correspondence	FLCR-03, FLCR-16, SMD-10	Closes addressability/bookkeeping analogy only.

This paper can now make the stronger claim that electron-hole-exciton formalism is an available standard model for vacancy, complement, recombination, screening, and bound-state behavior. LCR can use that formalism where it preserves the distinction between standard material model and internal residue address.

Active Comparator Rows

Comparator	External side	LCR side	Current result	Next required action
Hole/vacancy	unoccupied band state	open residue/vacancy slot	correspondence closed	Keep as model correspondence unless a material system is named.
Exciton bound state	electron-hole pair with effective mass/screening	paired residue/excitation lane	formalism closed	Add material parameters for quantitative claims.
Recombination/relaxation	material-dependent process	correction/repair path	analogy only	Requires dataset, rate equation, or simulation.

Executable Evidence Bound In This Pass

This paper now treats electron-hole-exciton formalism as a closed external vocabulary when material assumptions are stated, and as an open calibration row when material parameters are missing. That lets the paper stop treating holes and excitons as mysterious placeholders while still preserving the boundary around measured material behavior.

Evidence	Path	Result imported here	Boundary preserved
Electron-hole-exciton definition surface	SMD-10	Closes the standard vocabulary: hole as an unoccupied state in a filled band with positive effective carrier behavior; exciton as an electron-hole bound state modeled by screening and effective masses.	Does not close material-specific binding energy, recombination, transport, or ontology beyond the model.
Exciton source registry	local evidence artifact	Supplies IRL-EXCITON-SPRINGER and IRL-EXCITON-2D-REVIEW as citation-bound external sources.	External sources close definitions and formal targets, not FLCR-specific physical identity.
NSIT reasoned closure pass	local evidence artifact	States that electron-hole-exciton formalism closes a large part of the vacancy/complement vocabulary for papers 07, 15, 16, and 22.	Requires material model, Hamiltonian/equations, units, and citation/data before measured claims promote.

The upgraded claim is: FLCR-34 can use electron-hole-exciton formalism to explain vacancy/residue bookkeeping where the analogy is explicitly declared and the material model is named. The paper cannot claim material-specific exciton energies or recombination behavior without parameter rows.

Online Source Rows Applied In This Pass

Online source	Claim-lane effect	Boundary retained
IRL-EXCITON-2D-REVIEW	Closes standard electron-hole-exciton vocabulary: bound electron-hole pairs, reduced screening, stronger Coulomb interactions, and material-dependent exciton physics.	Material-specific binding and recombination remain parameter/data-bound.
IRL-NIST-CARRIER-DYNAMICS	Supplies experimental/metrology lane for exciton formation, electron-hole separation, recombination, and free-carrier dynamics.	Measurements require a chosen material system and protocol.
IRL-NIST-PV-EXCITON	Supplies materials row for exciton separation into free electrons and holes at heterojunction interfaces.	Does not turn LCR vacancy bookkeeping into a measured material claim.

Online data therefore closes the claim that electron-hole-exciton formalism is available and relevant. The remaining work is mapping a specific LCR residue to a specific material model and measured dataset.

Assertive Closure Promotions

Row	Closure now allowed	Evidence	Residue moved out of the open row
Hole vocabulary	closed by standard formalism	SMD-10, IRL-EXCITON-2D-REVIEW	Ontological claim beyond the model.
Exciton vocabulary	closed by standard formalism	SMD-10, exciton review, NIST carrier/PV sources	Material-specific binding, lifetime, recombination.
LCR vacancy/residue correspondence	closed as bookkeeping/addressability analogy when declared	NSIT_REASONED_CLOSURE_PASS, SMD-10	Measured material claim without parameters.

The non-timid conclusion is: the electron-hole-exciton bridge is not an open mystery. It is solved as a standard-formalism correspondence and remains data-bound only for specific materials.

Appendix D. Resolved-State Closure Pass

This pass removes false restrictions on the paper’s claim posture. A row is no longer called open merely because a stronger future claim exists. The satisfied lane is closed now; only the stronger claim that lacks its required adapter, comparator, calibration, or falsifier remains as residue.

Closed Now

Row	Lane	Resolved state	Remaining boundary
paper lift enumeration	receipt_bound_internal_result	8/8 LCR states succeeded; error walls=0; avg TarPit mass=0.513090.	This closes the paper-local finite lift readout, not every stronger physical interpretation.
representation fold	normal_form_result	The fold/transport action is closed as a representation map with named invariants and bounded residues.	The family closure is a structural lane; measured physics still requires destination-specific calibration.
decade-4 tower	receipt_bound_internal_result	The decade tower is resolved with avg TarPit mass=0.509077 and mass sum=40.726174.	The decade total is an internal tower metric, not a standalone universal physical constant.
family-04 cross-lift comparison	cam_crystal_reapplication_result	Family tower binds FLCR-04, FLCR-14, FLCR-24, FLCR-34; avg TarPit mass=0.508927; error walls=0.	Cross-lift agreement strengthens routing and comparison; it does not erase paper-local boundaries.
electron-hole-exciton formalism	standard_theorem_citation_bound_result	Electron-hole-exciton vocabulary is closed by standard condensed-matter formalism.	Material-specific exciton values require material parameters.

Claim Posture After This Pass

FLCR-34 should now be read as a resolved-state contribution for Electron-Hole-Exciton Accounting at its declared lane. The paper may state the strongest claim supported by these rows directly. It should reserve caution only for a specifically named stronger claim whose required evidence is absent.

8. Dependencies And Downstream Use

This paper may be imported by later FLCR papers only through the claim lanes above. The default downstream consumers are:

- FLCR-11 through FLCR-18 for window, receipt, admission, CA, algebraic, curvature, residue, and exceptional machinery.
- FLCR-28 through FLCR-30 for CAM/crystal, corpus ribbon, and universal normal-form intake.
- FLCR-31 through FLCR-40 only after the LCR-native result has been cited and the translation claim has its own evidence lane.

9. Limitations And Falsifiers

- material-specific bands, lifetimes, mobilities, and spectra require data-bound calibration

- External calibration claims require units, datasets, citations, and reproducible data binding.
- A later translation paper may strengthen this result only by adding the missing lane evidence.

A falsifier for this paper is any example that satisfies the declared input conditions while violating the stated construction, receipt, or lane boundary. An interpretation that merely wants a stronger downstream conclusion is not a falsifier; it is an obligation routed to a later paper.

10. Publication Readiness Tasks

- Validator identities and receipt hashes are recorded in the manifest or receipt appendix.
- External theorems require final citation entries before journal submission.
- Every theorem, proposition, and protocol must retain a claim lane and evidence boundary.
- The Markdown, LaTeX, and PDF forms are generated from the same canonical source.

Dual Positioning: Story And Formal Carrier

This paper must be read in two synchronized ways. Sequentially, it explains why the next state exists in the corpus story. Formally, it binds that state to accepted mathematics, IRL formalism, code receipts, validators, CAM/crystal lookups, and claim-lane boundaries.

Assigned original ribbon role(s): orbital publication overlay.

Original slot	Family	Lift	Role	Proof form
orbital	publication overlay	n/a	no legacy original slot assigned yet	intake and routing proof

The formal obligation is to state the strongest valid claim available for this paper without letting interpretation silently change the addressed object. Any author interpretation belongs beside the formalism as a declared relabeling, bridge, or analog, and must be accompanied by the evidence lane that makes the claim consumable by later papers.

State-Bound Proof Contract

This paper receives: mature LCR-native corpus and orbital evidence intake.

It must produce: Electron-Hole-Exciton Accounting as publication overlay or synthesis route.

Semantic transition: route external, comparative, or synthesis material around the original proof ribbon without overwriting original slot obligations.

Accepted formalism targets to bind in the journal rewrite:

- source routing and dependency graphs
- citation-bound comparison
- normal-form correspondence tables
- falsifier and calibration boundaries

Slot	Contract
orbital	publication overlay; must declare sources, dependencies, and calibration/falsifier boundaries

The formal carrier uses these accepted tools while preserving interpretive claims as explicit relabelings, correspondences, or bridges. Claim promotion is admissible only when a receipt, standard citation, CAM/crystal reapplication, normal-form result, calibration datum, or falsifier boundary is attached.

Provenance And Crystal Evidence Integration

This section integrates prior work, CAM/crystal projections, NSIT tooling, standards-window evidence, and routed supplements as provenance-bearing evidence. Source material is not treated as manuscript text; it is bound into the paper only through

claim lanes, receipts, validators, normal forms, calibration rows, or explicit falsifier boundaries.

Expansion Rule For This Slot

Use past work as orbital evidence: route all imports through dependency, citation, normal-form, calibration, and falsifier lanes.

Primary Evidence Inputs

Source	Placement reason
NP-12_Electron_Hole_Exciton_Accounting_For_Open_Math.md	primary assigned source for the paper's formal spine
supplements/SM_BRIDGE_SUPPLEMENT.md	primary assigned source for the paper's formal spine

Secondary And Orbital Evidence Inputs

Source	Placement reason
supplements/SM_BRIDGE_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/RECEIPT_VERIFIER_CATALOG.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_PYRAMID_INDEX.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_5P_WINDOW_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_7P_WINDOW_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_9P_WINDOW_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
CAM_CLAIM_100_SCORE_AUDIT.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
NSIT_TOOL_CLOSURE_MATRIX.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
NSIT_REASONED_CLOSURE_PASS.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
ORBITAL_DATA_CROSS_POPULATION_AUDIT.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
AGENT_CRYSTAL_WORK_HARVEST.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
PAPER_REWORKS_CRYSTAL_PROJECTION.json	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
Additional source routes	Complete routing is maintained in the source-placement appendix

Crystal And Standards Evidence To Bind

- Paper Reworks crystal projection: 33 paper markdown files, 9 supplements, 5 queues, 6 lattice-forge unification artifacts, 140 faces, 140 vignettes, and 280 CAM nodes.
- Standards-window intake: 192/192 corpus conformance verdicts PASS; 140/142 obligations have candidate routes; 2/4/8/16/32 window lattice is available for cross-paper reads.
- Agent/crystal harvest: agent-generated memory, runtime standards starter, current runtime build, repo-harvest CAM, notebook-derived bundles, and downloaded crystal payloads are orbital evidence surfaces.
- NSIT inventory baseline: 220 validators, 1786 receipt/data artifacts, 1137 formal-paper-like artifacts, 114 supplements, and 86 memory/CAM artifacts.

Paper-Specific CAM/Score Rows

No direct 00 - 32 CAM score row is assigned; use this paper as an orbital or translation intake.

Paper-Specific NSIT Closure Rows

No direct NSIT row matched the assigned legacy papers in the first-pass matrix; validator matching by object name remains a receipt-attachment requirement.

Source Integration Summary

The legacy source and internal reasoning supplement are treated as source evidence rather than manuscript prose. Their contributions enter this paper as claim-lane entries, receipt or validator references, finite-domain statements, and limitation boundaries. Speculative or cross-domain interpretations remain separate from closed local results until an explicit adapter, citation, normal-form derivation, calibration row, or falsifier is attached.

Claim Binding Queue

This queue records the evidence discipline for claim strengthening. It separates claims that already have support from residues that remain in falsifier or open-obligation lanes.

Queue	Promote now	Bind next	Residue
Electron-Hole-Exciton Standard Accounting	Promote standard hole/exciton vocabulary and LCR vacancy/residue bookkeeping when SMD-10 is cited.	Bind material model, band structure, effective masses, screening, and parameter rows before material-specific claims.	Binding energy, recombination, lifetime, and transport stay calibration/data-bound.

Binding Discipline

Each queue item is represented as a delta:

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local claim at paper time
-> attached later source/tool/receipt/theorem
-> revised representation
-> invariant boundary and remaining residue
```

This preserves the sequential story while allowing later tools, crystals, and standard formalisms to return through the proper lane.

Claim Delta Ledger

This ledger applies the paper's claim-binding queues as explicit back-application deltas. It keeps the sequential paper story intact while allowing later receipts, standard formalisms, CAM/crystal routes, and validators to sharpen the claim.

Delta	Local claim at paper time	Later evidence	Revised representation	Boundary kept
Search By Object Name	No direct reasoned-closure queue was assigned from the first queue map.	Search source routing, verifier catalog, CAM/crystal routes, and paper-local object names.	Create a specific binding queue when an object-name match finds a validator, receipt, theorem, or calibration route.	Until then, claims remain in their existing lanes.

The revised representation records the strongest claim supported by the current evidence package. Promotion into theorem or proposition language occurs only when the named evidence is attached in the manifest or workbook replay.

Source-Backed Evidence Summary

Routed archive and mirror cards are treated as provenance summaries rather than body text. They identify candidate receipts, validators, theorem registries, or supplemental source routes. A card strengthens this paper only when its contribution is bound to a claim lane, evidence object, and boundary.

Evidence card	Score	Publication contribution	Boundary
No routed archive card attached yet	0	source-backed contribution	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
Additional evidence cards	0 total	The complete card inventory is maintained in the archive evidence matrix.	Cards are source-discovery aids until bound to paper-local evidence.

Journal Apparatus

Article Type And Intended Venue Fit

This manuscript is written as a formal-methods and mathematical-systems paper inside the series *Formalizing LCR, Unifying Standard Models*. Its intended reader is a technical reviewer who needs claim boundaries, reproducibility routes, and cross-paper dependencies to be visible without relying on private context.

Structured Contribution Statement

Item	Statement
Publication ID	FLCR-34
Legacy source slot(s)	orbital
Ribbon role	routes external, standard-model, or synthesis material around the original proof ribbon
Proof form	source intake, correspondence table, dependency statement, and falsifier boundary
Received state	mature LCR-native corpus and orbital evidence intake
Produced state	Electron-Hole-Exciton Accounting as publication overlay or synthesis route

Claim-Evidence Table

Claim	Lane	Evidence to attach	Boundary
Theorem 34.1	standard_theorem_citation_bound_result	Receipt, source card, validator, citation, or CAM route named in the paper manifest	standard electron, hole, recombination, screening, and exciton models can explain part of the open residue surface when material context is declared
Proposition 34.2	normal_form_result	Receipt, source card, validator, citation, or CAM route named in the paper manifest	exciton accounting can be imported by later papers only through its declared source, receipt, and lane.
Protocol 34.3	falsifier_or_open_obligation	Receipt, source card, validator, citation, or CAM route named in the paper manifest	material-specific bands, lifetimes, mobilities, and spectra require data-bound calibration

Figure Plan

Figure	Purpose	Caption
FLCR-34-F1	Publication overlay dependency map	Schematic showing how FLCR-34 turns its received state into the produced state while preserving claim lanes and residue boundaries.
FLCR-34-F2	Evidence routing map	Diagram of source papers, archive cards, CAM/crystal routes, validators, and workbook replay paths used by this manuscript.
FLCR-34-F3	Same-family lift context	Diagram placing this paper in the original 00-79 ribbon family and showing predecessor/successor slots.

In-Text Figure FLCR-34-F1: Publication overlay dependency map

```
received state
-> Publication overlay dependency map
-> formal claim lane
-> evidence object / receipt / citation
-> produced state
-> remaining residue
```

Table Plan

Table	Purpose
FLCR-34-T1	Dependency and translation table
FLCR-34-T2	Claim-lane/evidence-boundary table
FLCR-34-T3	Archive evidence card and source-backed upgrade table
FLCR-34-T4	Workbook replay and falsifier table

Worked Example

Route one standard-model or synthesis claim back to its LCR-native source papers. The example must name the input object, the operation, the evidence object, the allowed revised claim, and the remaining boundary.

Nomenclature And Glossary

Term	Paper-local meaning
CAM	Defined by this paper as part of the publication_overlay proof role; final definition is recorded in the paper-local glossary.
claim lane	Defined by this paper as part of the publication_overlay proof role; final definition is recorded in the paper-local glossary.
crystal	Defined by this paper as part of the publication_overlay proof role; final definition is recorded in the paper-local glossary.
dependency	Defined by this paper as part of the publication_overlay proof role; final definition is recorded in the paper-local glossary.
publication overlay	Defined by this paper as part of the publication_overlay proof role; final definition is recorded in the paper-local glossary.
receipt	Defined by this paper as part of the publication_overlay proof role; final definition is recorded in the paper-local glossary.
synthesis lane	Defined by this paper as part of the publication_overlay proof role; final definition is recorded in the paper-local glossary.
translation claim	Defined by this paper as part of the publication_overlay proof role; final definition is recorded in the paper-local glossary.

Appendix FLCR-34-A: Evidence Package

The appendix records:

1. Legacy source excerpts or source-card references used in the final claim ledger.
2. Receipt and verifier paths, including pass/fail/partial status.
3. CAM/crystal query or projection rows used by the paper, with source identity and boundary.
4. Standards-window and NSIT evidence used for conformance or routing.
5. Falsifier, calibration, or external-data rows that remain unresolved.

Data And Code Availability

The publication package contains the canonical Markdown sources, manifests, source-routing matrices, validation reports, and generated PDF/TeX outputs.

Code and verifier references are cited through manifests, archive evidence cards, receipt catalogs, or workbook replay tables. External datasets or standards citations must be attached before any calibration-bound claim is promoted.

Reviewer Checklist

Check	Reviewer question
Claim lane	Does every strong claim declare its lane and evidence type?
Boundary	Does the paper preserve what it does not prove?
Reproducibility	Is there a receipt, verifier, source card, citation, or replay table for the claim?
Cross-paper import	Does the paper cite the prior FLCR/OPR slot before consuming its result?
Interpretation	Does the analog layer preserve the addressed state while changing labels/access paths?

Declarations

No human-subjects, clinical, or animal-research protocol is asserted by this paper. External empirical claims require separate dataset, calibration, or domain-validation attachments. Competing-interest and funding statements are recorded in the submission metadata.

11. Publication Integration Addendum

11.1 Role In The Series

FLCR-34 (Electron-Hole-Exciton Accounting) occupies the publication_overlay role at lift depth n/a. The paper receives mature LCR-native corpus and orbital evidence intake. It produces Electron-Hole-Exciton Accounting as publication overlay or synthesis route. The state transition is: route external, comparative, or synthesis material around the original proof ribbon without overwriting original slot obligations.

The paper-local proof form is:

source intake, correspondence table, dependency statement, and falsifier boundary

The integration result is a source-intake and synthesis surface with dependency, citation, calibration, and falsifier boundaries. This addendum records how that result is consumed by the publication series without relying on editorial instructions or private working context.

11.2 Formal Carrier

Formal carrier	Function
source routing and dependency graphs	Formal carrier for the paper-local result.
citation-bound comparison	Formal carrier for the paper-local result.
normal-form correspondence tables	Formal carrier for the paper-local result.
falsifier and calibration boundaries	Formal carrier for the paper-local result.

The LCR, CAM, crystal, forge, and analog vocabulary functions as a typed access layer over these carriers. A relabeling is admissible only when the addressed object, evidence lane, boundary, and downstream import rule remain attached.

11.3 Claim Ledger

Claim	Lane	Statement
Theorem 34.1	standard_theorem_citation_bound_result	standard electron, hole, recombination, screening, and exciton models can explain part of the open residue surface when material context is declared
Proposition 34.2	normal_form_result	exciton accounting can be imported by later papers only through its declared source, receipt, and lane.
Protocol 34.3	falsifier_or_open_obligation	material-specific bands, lifetimes, mobilities, and spectra require data-bound calibration

These claims are consumed at their stated lane. A stronger downstream statement creates a new claim envelope with its own evidence object.

11.4 Evidence Package

Evidence class	Routed items	Status
Legacy sources	NP-12_Electron_Hole_Exciton_Accounting_F or_Open_Math.md, supplements/SM_BRIDGE_SUPPLEMENT.md	routed evidence
Evidence inputs	CAM_CLAIM_100_SCORE_AUDIT.md, NSIT_TOOL_CLOSURE_MATRIX.md, NSIT_REASONED_CLOSURE_PASS.md, PAPER_REWORKS_CRYSTAL_PROJECTION.json	routed evidence
CAM/crystal sources	PAPER_REWORKS_CRYSTAL_PROJECTION.json, AGENT_CRYSTAL_WORK_HARVEST.json, runtime standards_artifact, notebook-derived source bundle	routed evidence
Standards-window inputs	window_summary.json, node DBs, window DBs	routed evidence
Receipts	external requirement	not attached in this source package
Validators	external requirement	not attached in this source package

Standards-window, runtime, and notebook-derived artifacts are treated as evidence-routing surfaces. They support source discovery, conformance checking, and reapplication, but they do not replace citations, receipts, normal-form derivations,

calibration rows, or falsifiers.

11.5 Results Representation

The paper's results section is organized around five publication objects:

- 1. the exact object acted on at this slot;
- 2. the admissible operation or transformation;
- 3. the receipt, enumeration, derivation, lookup, or external theorem supporting the operation;
- 4. the residue or boundary left after the result;
- 5. the downstream import rule.

11.6 Figures And Tables

Figure	Role
FLCR-34-F1: Publication overlay dependency map	Visualizes the proof object, routing, or boundary.
FLCR-34-F2: Evidence routing map	Visualizes the proof object, routing, or boundary.
FLCR-34-F3: Same-family lift context	Visualizes the proof object, routing, or boundary.

Table	Role
FLCR-34-T1: Dependency and translation table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-34-T2: Claim-lane/evidence-boundary table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-34-T3: Archive evidence card and source-backed upgrade table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-34-T4: Workbook replay and falsifier table	Records claim lanes, evidence, sources, or falsifiers.

11.7 Limits And Falsifiers

The paper proves only the object and domain declared in its claim ledger. Physical, Standard Model, observer, calibration, or synthesis claims require the additional lane evidence stated by their destination papers. CAM/crystal and forge language is operational only when represented as an address, lookup, receipt, validator, or reapplication path.